Geology Rocks!

Rocks are made of minerals, and every rock has a story of how it formed. You will test minerals like a scientist, learn to read the clues in rocks, and follow Earth's materials around the rock cycle.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 2	Rock or Mineral? Be a Rock Detective
Lesson 2	Fri Nov 6	Mineral Tests: Hardness, Streak, Luster, Cleavage
Lesson 3	Fri Dec 11	Metallic Minerals, Look-Alikes, and Mineral Uses
Lesson 4	Fri Jan 29	Igneous, Sedimentary, and Metamorphic Rocks
Lesson 5	Fri Feb 26	Practice Tournament
Lab I1	Before Lesson 3	Kitchen Mineral Lab: Scratch, Streak, and Fizz (home or an open Friday)
Lab I2	Before Lesson 4	Rock Cycle Journey and Squish Lab (home or an open Friday)

At the tournament

A 50-minute station event: partners rotate through timed stations to identify real rock and mineral specimens from the official list, classify them by their properties, and answer questions about uses and the rock cycle, bringing pencils, up to 2 hand lenses, and one 1-inch binder of notes.

Keep every handout for this event behind this divider, in order. Bring this binder to every Field A Fridays meeting.

Geology Rocks!: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can tell a rock from a mineral and give the 5 parts of what makes a mineral.	L1	☐	☐
I can do a hardness test and use the Mohs scale (talc 1 to diamond 10) with a fingernail (2.5), copper penny (about 3.5), glass (5.5), and steel file (6.5).	L2 L3 L5 I1	☐	☐
I can do a streak test and use streak color to tell look-alikes apart (gold vs pyrite, hematite).	L2 L3 L5 I1	☐	☐
I can describe luster (metallic, glassy, pearly, greasy, dull) and tell cleavage from fracture.	L2 L3	☐	☐
I can use special clues: fizz in acid (calcite), cubes that dissolve (halite), thin sheets (mica), very heavy (galena, gold), soapy or greasy (talc, graphite), glow (fluorite).	L2 L3 I1	☐	☐
I can identify all 16 minerals on the list: calcite, copper, corundum, feldspar (pink), fluorite, galena, gold, graphite, gypsum, halite, hematite, mica, olivine, pyrite, quartz (crystal), and talc.	L2 L3 L5	☐	☐
I can name a use for each mineral and explain why minerals are important to the economy.	L3 L5 I1	☐	☐
I can draw and explain the rock cycle: weathering, erosion, deposition, compaction, cementation, melting, cooling, and heat and pressure.	L5 I2	☐	☐
I can identify the 5 igneous rocks (basalt, granite, obsidian, pumice, scoria) and explain how cooling speed changes crystal size.	L4 L5	☐	☐
I can identify the 6 sedimentary rocks (bituminous coal, conglomerate, chemical limestone, fossil limestone, sandstone, shale) and tell how each formed.	L4 L5	☐	☐
I can identify the 6 metamorphic rocks (gneiss, marble, phyllite, quartzite, schist, slate), tell foliated from nonfoliated, and name each parent rock.	L4 L5 I2	☐	☐
I can classify a rock as igneous, sedimentary, or metamorphic from clues like crystals, grains, layers, bands, holes, and fossils.	L1 L4	☐	☐
I can name uses of rocks (buildings, countertops, statues, roofs, fuel, cement).	L4	☐	☐
I can find any specimen in our 1-inch binder in under 30 seconds.	L2 L5	☐	☐

Rock or Mineral? Be a Rock Detective

TODAY I CAN

- I can tell a rock from a mineral and give the 5 parts of what makes a mineral.
- I can classify a rock as igneous, sedimentary, or metamorphic from clues like crystals, grains, layers, bands, holes, and fossils.

Geologists are scientists who study rocks. This is page 1 of your team's 1-inch Geology binder.

Today's steps: run it yourselves (22 minutes)

You need: Kate's quartz, pyrite, calcite, obsidian (and the mica piece if it peels), Numbered mystery rocks from the coach's bag, 2 pennies and 2 steel nails, Cup of kitchen salt, cup of vinegar with a dropper or spoon, paper towel, 4 station cards (1-4), Lesson 1 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Hold each of Kate's minerals: quartz, pyrite, calcite. Each one is ONE ingredient. Now use a hand lens on a mystery rock. Can you find specks of different colors? A rock is like a cookie; minerals are the ingredients.
2. Both 2 min. Set the timer for 2 minutes. Take turns reading the 'Rock or mineral?' box out loud. Say the 5 parts of a mineral to your partner without looking.
3. Both 7 min. Set the timer for 7 minutes. Take 2 numbered mystery rocks. Fill in the 'Mystery Rocks' table. Scratch test: fingernail, then penny, then steel nail; rub off the powder to see if it left a real groove. Nails point down at the table.
4. Event Judge 6 min. Put the 4 station cards and items around the table. Set the timer for 90 seconds per station and call 'Rotate!'. Pairs fill in the 'Station Sprint' table and cannot go back.
5. Both 3 min. Set the timer for 3 minutes. Secret sort: one pair sorts all the mystery rocks into 2 piles by one clue (shiny vs dull, crystals vs grains). The other pair guesses the rule. Switch.
6. Both 1 min. Answer the 'Think' question. Check your stations with the answer card and file the page.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Rock or mineral?

Mineral	1) formed in nature, 2) not alive, 3) solid, 4) has its own chemical recipe, 5) atoms lined up in a crystal pattern. Examples: quartz, halite (salt).
Rock	a solid mix of one or more minerals (or volcanic glass, or old plant bits). Example: granite = quartz + feldspar + mica.
Luster	how it shines: metallic, glassy, shiny, or dull
Scratch test	try fingernail, then penny, then steel nail. The first one that leaves a groove tells how soft it is.

Mystery Rocks

Rock #	Color	Luster	What I see (crystals, grains, layers, holes, glassy)	Scratched by	One mineral or a mix?

Station Sprint

Station	Rock or mineral?	Clue
1. Salt: what shape are the grains?		
2. Gold or not? Try the nail.		
3. Vinegar drop: what happens?		
4. Why no crystals?		

Think

1. In your own words, what is the difference between a rock and a mineral?

At home before next time: Rock and mineral hunt at home (15 minutes). With an adult, find 5 things that come from rocks or minerals. Ideas: salt (halite), pencil (graphite), baby powder (talc), walls (gypsum drywall), a countertop (granite), a copper wire, jewelry. Write each item and its rock or mineral on the Minerals at Home page of your Rock Detective Field Journal (p. 15). That earns the Mineral Spotter badge. Optional: watch the 'What Is a Mineral?' video on the NCSO page.

Mineral Tests: Hardness, Streak, Luster, Cleavage

TODAY I CAN

- I can do a hardness test and use the Mohs scale (talc 1 to diamond 10) with a fingernail (2.5), copper penny (about 3.5), glass (5.5), and steel file (6.5).
- I can do a streak test and use streak color to tell look-alikes apart (gold vs pyrite, hematite).
- I can describe luster (metallic, glassy, pearly, greasy, dull) and tell cleavage from fracture.
- I can use special clues: fizz in acid (calcite), cubes that dissolve (halite), thin sheets (mica), very heavy (galena, gold), soapy or greasy (talc, graphite), glow (fluorite).
- I can identify all 16 minerals on the list: calcite, copper, corundum, feldspar (pink), fluorite, galena, gold, graphite, gypsum, halite, hematite, mica, olivine, pyrite, quartz (crystal), and talc.
- I can find any specimen in our 1-inch binder in under 30 seconds.

Minerals can be identified by tests called properties. This page goes in your Geology binder.

Today's steps: run it yourselves (22 minutes)

You need: 9 test minerals (talc, gypsum, calcite, fluorite, halite, mica if it peels, feldspar, quartz, corundum), Apatite, topaz, and silicon carbide from the Mohs set (ladder only), 2 streak plates, 2 taped glass plates, 2 nails, 2 pennies, 10 Mohs ladder cards, Lesson 2 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Take turns reading 'The 4 big tests' out loud. Streak color can be different from the outside color.
2. Both 3 min. Set the timer for 3 minutes. Practice on calcite: rub it on the streak plate, try to scratch it with a fingernail and a penny, then try to scratch the glass with it (glass stays flat on the table). Wipe: a real scratch stays.
3. Both 9 min. Set the timer for 9 minutes. Pair 1 tests talc, gypsum, calcite, fluorite, and halite. Pair 2 tests mica (only if it peels into sheets), feldspar, quartz, and corundum. Fill in your rows of the 'Test lab' table. Scratch gently with short strokes.
4. Both 3 min. Set the timer for 3 minutes. Lay the 10 Mohs cards in a line and put the real minerals from the Mohs set on 1 to 9. Diamond (10) is a card only. Say: Tall Giraffes Can Fight Anyone For Quite Tough Cookies Daily. Answer 'Check yourself' question 1.
5. Team 2 min. Set the timer for 2 minutes. Share results with the other pair so every row is filled. Answer questions 2 and 3.
6. Event Captain 2 min. Read the answers from the answer card. Pairs fix mistakes and file the page.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

The 4 big tests

Hardness	how hard it is to scratch. Mohs scale: 1 talc, 2 gypsum, 3 calcite, 4 fluorite, 5 apatite, 6 feldspar, 7 quartz, 8 topaz, 9 corundum, 10 diamond.
Test tools	fingernail 2.5, copper penny about 3.5, glass 5.5, steel nail or file about 6.5
Streak	the color of the mineral's powder when rubbed on a white tile
Luster	how it shines: metallic (like metal), glassy, pearly, greasy, or dull
Cleavage	breaks along flat, smooth surfaces. Fracture: breaks unevenly or in curved shell shapes.

Test lab

Mineral	Hardness (scratched by?)	Streak	Luster	Cleavage or fracture
Talc				
Gypsum				
Calcite				

Mineral	Hardness (scratched by?)	Streak	Luster	Cleavage or fracture
Fluorite				
Halite				
Mica				
Feldspar (pink)				
Quartz (crystal)				
Corundum				

Check yourself

1. Write the Mohs scale from 1 to 10.

2. What is the difference between cleavage and fracture?

3. A mineral scratches glass, but quartz scratches it. About how hard is it? Which listed mineral could it be?

At home before next time: Do Independent Lab I1 (Kitchen Mineral Lab) before Lesson 3 on Dec 11. It uses the tests you learned today. Record your results in the Scratch and Streak Lab in your Field Journal (p. 8) for the Scratch Tester badge. Also practice the Mohs memory line (journal p. 3) until you can say all 10 minerals in order. Museum trip tomorrow (Sat Nov 7, meet 10 am at the Daily Planet Cafe): bring your journal, find 3 minerals from our list, and fill in the Museum Day page (p. 17).

Metallic Minerals, Look-Alikes, and Mineral Uses

TODAY I CAN

- I can do a hardness test and use the Mohs scale (talc 1 to diamond 10) with a fingernail (2.5), copper penny (about 3.5), glass (5.5), and steel file (6.5).
- I can do a streak test and use streak color to tell look-alikes apart (gold vs pyrite, hematite).
- I can describe luster (metallic, glassy, pearly, greasy, dull) and tell cleavage from fracture.
- I can use special clues: fizz in acid (calcite), cubes that dissolve (halite), thin sheets (mica), very heavy (galena, gold), soapy or greasy (talc, graphite), glow (fluorite).
- I can identify all 16 minerals on the list: calcite, copper, corundum, feldspar (pink), fluorite, galena, gold, graphite, gypsum, halite, hematite, mica, olivine, pyrite, quartz (crystal), and talc.
- I can name a use for each mineral and explain why minerals are important to the economy.

Mineral Master Chart. Keep this in the front of your Minerals tab; it is the page you will use most at the tournament.

Today's steps: run it yourselves (27 minutes)

You need: All 16 minerals: real pieces, pencil lead, copper wire, and photo cards (gold, galena, hematite, olivine; mica if needed), 2 streak plates, 2 taped glass plates, 2 nails, 2 pennies, vinegar cup and dropper, 4 mystery cups A-D, 16 use cards, Lesson 3 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Read one row of the 'Mineral Master Chart' out loud together. Then read 'ID tips'. For metallic minerals, streak is the best test.
2. Both 8 min. Set the timer for 8 minutes. Streak-test pyrite and read the gold and hematite cards. Which is fool's gold? Compare graphite (rub the pencil lead on paper) with the galena card. Bend the copper wire and read the olivine card. Answer 'Check yourself' questions 1 and 2.
3. Both 7 min. Set the timer for 7 minutes. Identify mystery cups A-D with tests and the chart (vinegar is for 1 drop only). Write the mineral and the test that proved it in 'Mineral mystery'.
4. Both 6 min. Set the timer for 6 minutes. Deal the 16 use cards and match each one to its mineral. Talk: why do people need minerals? Answer question 3.
5. Event Captain 3 min. Read the answers from the answer card. Pairs fix mistakes, wipe the streak plates, and file the chart behind the Minerals tab.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Mineral Master Chart (softest to hardest)

Mineral	Hardness	Streak	Luster	Cleavage or fracture	Clues and uses
Talc	1	white	pearly or greasy	1 direction	softest mineral; feels soapy. Baby powder, paint, soapstone.
Graphite	1-2	black or dark gray	metallic to dull	1 direction	greasy, marks paper. Pencil 'lead', lubricant, batteries.
Gypsum	2	white	glassy to pearly or silky	1 direction	fingernail scratches it. Drywall, plaster of Paris.
Mica	2-2.5	white	glassy to pearly	1 perfect: peels in thin sheets	flexible sheets. Electrical insulation, makeup shimmer. Mined in NC.
Halite	2.5	white	glassy	3 directions at right angles (cubes)	dissolves in water. Table salt, road salt.
Galena	2.5	lead gray	metallic	3 directions at right angles (cubes)	very heavy, silver-gray cubes. Main ore of lead.
Gold	2.5-3	golden yellow	metallic	none	very heavy, bends without breaking. Jewelry, electronics, money.
Copper	2.5-3	copper red	metallic	none	copper color, may have green crust. Wires, pipes, coins.
Calcite	3	white	glassy	3 directions, not at right angles (rhombs)	fizzes in acid; clear pieces show double images. Cement, antacids.

Mineral	Hardness	Streak	Luster	Cleavage or fracture	Clues and uses
Fluorite	4	white	glassy	4 directions (8-sided pieces)	purple, green, or blue cubes; may glow under black light. Steel making, fluoride.
Hematite	5.5-6.5	red-brown	metallic or dull	none	red-brown streak even when silver-black. Main ore of iron.
Feldspar (pink)	6	white	glassy	2 directions at right angles	pink, blocky; in granite. Ceramics, glass.
Pyrite	6-6.5	greenish black	metallic	none	brassy cubes; 'fool's gold'. Source of sulfur.
Olivine	6.5-7	white	glassy	poor; breaks unevenly	olive green grains, often in basalt. Gem called peridot.
Quartz (crystal)	7	white	glassy	none; curved shell-like fracture	6-sided crystals, scratches glass. Glass, electronics, sand. NC quartz is used for computer chips.
Corundum	9	white	glassy to shiny (adamantine)	none	very hard 6-sided barrels. Rubies and sapphires, sandpaper.

ID tips

Metallic?	Then use streak: yellow = gold, greenish black = pyrite, red-brown = hematite, gray = galena, black and greasy = graphite, copper-red = copper.
Not metallic?	Then use hardness and cleavage: sheets = mica, cubes = halite, rhombs and fizz = calcite, 8-sided = fluorite, 6-sided crystal = quartz, pink blocks = feldspar, green grains = olivine.

Mineral mystery

Cup	Mineral	Test that proved it
A		
B		
C		
D		

Check yourself

1. How can you tell pyrite from gold?

2. Which silver-black mineral has a red-brown streak?

3. Name 3 minerals and something each is used for.

At home before next time: Do Lab I2 (Rock Cycle Journey) before Lesson 4 on Jan 29. Then label the rock cycle in your Field Journal (p. 7) for the Rock Cycle Rider badge. Also make binder pages for your 8 of the 16 minerals (partner does the other 8): photo or drawing, hardness, streak, luster, cleavage, and one use. No real samples.

Igneous, Sedimentary, and Metamorphic Rocks

TODAY I CAN

- I can identify the 5 igneous rocks (basalt, granite, obsidian, pumice, scoria) and explain how cooling speed changes crystal size.
- I can identify the 6 sedimentary rocks (bituminous coal, conglomerate, chemical limestone, fossil limestone, sandstone, shale) and tell how each formed.
- I can identify the 6 metamorphic rocks (gneiss, marble, phyllite, quartzite, schist, slate), tell foliated from nonfoliated, and name each parent rock.
- I can classify a rock as igneous, sedimentary, or metamorphic from clues like crystals, grains, layers, bands, holes, and fossils.
- I can name uses of rocks (buildings, countertops, statues, roofs, fuel, cement).

There are 17 rocks on the list, in 3 families. Metamorphic (met-a-MOR-fik) means changed form. These pages go behind your Rocks tab.

Today's steps: run it yourselves (27 minutes)

You need: 14 real rocks and 4 photo cards (scoria, coal, fossil limestone, phyllite) in 3 trays; rhyolite bonus; calcite, 2 cups for water, vinegar cup, 2 droppers, paper towels, 1 taped glass plate, Lesson 4 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Take turns reading 'Rock clues' out loud. Say one clue for each family without looking.
2. Both 7 min. Set the timer for 7 minutes. Look at the 4 igneous rocks and the Scoria card with a hand lens: big crystals, tiny ones, glass, or holes? Fill a cup with water and drop in the pumice. Does it float? Read the Scoria card. Fill in the igneous rows of 'The 17 rocks'.
3. Both 7 min. Set the timer for 7 minutes. Sort the sedimentary rocks by grain size (pebbles, sand, mud). Put one drop of vinegar on the tufa, the plain limestone, and the calcite, and look for tiny bubbles. Read the Fossil limestone and Bituminous coal cards: coal comes from old swamp plants. Fill in the sedimentary rows.
4. Both 7 min. Set the timer for 7 minutes. Sort the metamorphic rocks and the Phyllite card into foliated and nonfoliated. Line up the shale line. Put a vinegar drop on marble and quartzite, and try to scratch the glass (flat on the table) with quartzite. Fill in the metamorphic rows.
5. Event Captain 3 min. Pairs answer 'Check yourself'. Read the answers from the answer card. Pairs fix mistakes, wipe and dry the rocks, and file the pages behind the Rocks tab.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Rock clues

Igneous	cooled from magma or lava. Slow cooling = big crystals, fast cooling = tiny crystals or glass, gas bubbles = holes.
Sedimentary	bits of rock, shells, or plants pressed and glued together. Look for grains, layers, or fossils. Limestone fizzes in vinegar.
Metamorphic	a rock changed by heat and pressure without melting. The parent rock is the rock it was before.
Foliated	minerals lined up in sheets, stripes, or sparkly layers (slate, phyllite, schist, gneiss)
Nonfoliated	no layers; crystals fit together like sugar (marble, quartzite)
The shale line	shale, then slate, then phyllite, then schist, then gneiss: more heat and pressure each step

The 17 rocks

Rock	Type	Clues (look and feel)	How it formed or parent rock / use
Basalt	igneous		
Granite	igneous		
Obsidian	igneous		
Pumice	igneous		
Scoria	igneous		
Bituminous coal	sedimentary		
Conglomerate	sedimentary		
Limestone (chemical)	sedimentary		
Limestone (fossil)	sedimentary		
Sandstone	sedimentary		
Shale	sedimentary		
Slate	metamorphic		
Phyllite	metamorphic		
Schist	metamorphic		
Gneiss	metamorphic		
Marble	metamorphic		
Quartzite	metamorphic		

Check yourself

1. What decides whether an igneous rock has big crystals, tiny crystals, or none?

---

2. Why does pumice float but scoria usually sinks?

---

3. List the shale line in order from least to most changed.

---

4. How can you tell marble from quartzite?

At home before next time: Finish your binder (15-20 minutes). Split the 17 rocks with your partner and make one binder page for each of yours: a photo or drawing, type, 2 clues, how it formed (or its parent rock), and a use. Check that the binder has tabs (Tests, Minerals, Rocks, Rock Cycle), a table of contents, every page attached, and no real samples. Then have someone call out 10 specimen names while you time how fast you find each page. Goal: under 30 seconds each. Field Journal: collect 3 rocks at home, test them, and sort them into families on Specimen pages (p. 9-14) for the Rock Hound badge. Bonus: take a Rock Walk (p. 16).

Practice Tournament

TODAY I CAN

- I can do a hardness test and use the Mohs scale (talc 1 to diamond 10) with a fingernail (2.5), copper penny (about 3.5), glass (5.5), and steel file (6.5).
- I can do a streak test and use streak color to tell look-alikes apart (gold vs pyrite, hematite).
- I can identify all 16 minerals on the list: calcite, copper, corundum, feldspar (pink), fluorite, galena, gold, graphite, gypsum, halite, hematite, mica, olivine, pyrite, quartz (crystal), and talc.
- I can name a use for each mineral and explain why minerals are important to the economy.
- I can draw and explain the rock cycle: weathering, erosion, deposition, compaction, cementation, melting, cooling, and heat and pressure.
- I can identify the 5 igneous rocks (basalt, granite, obsidian, pumice, scoria) and explain how cooling speed changes crystal size.
- I can identify the 6 sedimentary rocks (bituminous coal, conglomerate, chemical limestone, fossil limestone, sandstone, shale) and tell how each formed.
- I can identify the 6 metamorphic rocks (gneiss, marble, phyllite, quartzite, schist, slate), tell foliated from nonfoliated, and name each parent rock.
- I can find any specimen in our 1-inch binder in under 30 seconds.

Practice tournament! At each station, write the specimen name first, then answer the rest. When time is called, move on.

Today's steps: run it yourselves (27 minutes)

You need: 8 station specimens or photo cards and question cards (1-8), Streak plate, glass plate, penny, nail, vinegar and dropper, Lesson 5 answer card. Set the timer for each step.

1. Event Judge 1 min. Set out the 8 stations in order with their tools. Read aloud: '3 minutes per station. Move to the next number when time is called. Use your binder. Answer what you know first.' Pairs start at Stations 1 and 5.
2. Event Judge 12 min. Set the timer for 3 minutes each round for 4 rounds. Call 'One minute left' and 'Rotate!'. Write the name first, then the other answers, on the 'Answer sheet'.
3. Event Judge 12 min. Keep going for 4 more rounds so every pair visits all 8 stations. Leave specimens at their station.
4. Event Captain 2 min. Read the answers from the answer card. Pairs score their sheet and fill in 'My review plan'.

Check your work: Ask your Event Captain for the answer card and score your answer sheet. Circle 2 specimens to review and write them in 'My review plan'.

Answer sheet

Station	Name	Other answers
1		
2		
3		
4		
5		
6		
7		
8		

Before the tournament

- [ ] I can name all 17 rocks and 16 minerals on the list.
- [ ] I can do hardness, streak, luster, and cleavage tests.
- [ ] I can explain the rock cycle.
- [ ] My binder is 1 inch, pages are attached, and there are no real samples in it.

☐ I bring pencils and 2 hand lenses.

My review plan

1. Which 2 specimens or topics will I review?

At home before next time: Study 10 minutes every other day until the tournament on Sat Mar 13. Use the photo pages in your binder like flashcards: cover the name, look at the picture and clues, and guess. Focus on the look-alikes: pyrite and gold, marble and quartzite, slate and shale, pumice and scoria. Field Journal: cover the names on the Mineral and Rock Collector pages (p. 5-6) and name 10 specimens in under 2 minutes for the Speed Spotter badge.

Lab I1: Kitchen Mineral Lab: Scratch, Streak, and Fizz

THIS LAB HELPS ME

- I can do a hardness test and use the Mohs scale (talc 1 to diamond 10) with a fingernail (2.5), copper penny (about 3.5), glass (5.5), and steel file (6.5).
- I can do a streak test and use streak color to tell look-alikes apart (gold vs pyrite, hematite).
- I can use special clues: fizz in acid (calcite), cubes that dissolve (halite), thin sheets (mica), very heavy (galena, gold), soapy or greasy (talc, graphite), glow (fluorite).
- I can name a use for each mineral and explain why minerals are important to the economy.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Minerals are hiding in your kitchen. Test real things with the same tests geologists use. This page goes behind your Tests tab.

Steps (30 minutes)

You need: Eggshell or seashell, pencil, crayon, soap, salt, outside rock, Penny, cup of vinegar, spoon, paper towel, Streak plate (or the unglazed bottom of a mug), Hand lens. Set the timer for each step.

1. Me 3 min. Set a timer for 3 minutes. Read 'Test rules'. Line up your 6 test items on a paper towel in the same order as the 'Kitchen test table'.
2. Me 8 min. Set a timer for 8 minutes. Scratch test: try to scratch each item with your fingernail, then with the edge of the penny. Rub the spot with your finger; a real scratch stays. Write yes or no in the fingernail and penny columns.
3. Me 5 min. Set a timer for 5 minutes. Streak test: rub the pencil tip, the eggshell, a few salt grains, and the outside rock hard on the streak plate. Write the powder color in the Streak column.
4. Me 5 min. Set a timer for 5 minutes. Fizz test: drop a piece of eggshell into the cup of vinegar and watch with a hand lens for tiny bubbles. Next, put a few salt grains on the paper towel and add a spoonful of vinegar. Then do the same on the outside rock. Write fizz or no fizz.
5. Me 5 min. Set a timer for 5 minutes. Answer the 'Think like a geologist' questions. Then draw lines in 'Kitchen minerals'.
6. Me 4 min. Check your page with the Lab I1 answer card in the Open Lab bin (or ask your Event Captain at Lesson 3). Fix mistakes in a different color. Pour the vinegar down the sink and wash your hands.

Check your work: Use the Lab I1 answer card from the Open Lab bin, or bring the page to Lesson 3 and check it with your Event Captain. Your results can differ a little from the card; the big ideas should match.

Test rules

Hardness	Your fingernail is 2.5 and a penny is about 3.5. If your nail scratches it, it is softer than 2.5. If only the penny scratches it, it is about 3.
Streak	the color of the powder. Use only the rough, unglazed part of the plate or mug.
Fizz	Vinegar is a weak acid. Things made of calcium carbonate, like the mineral calcite, fizz with tiny bubbles.
Safety	No tasting. Do all tests over a paper towel. Wash your hands when you finish.

Kitchen test table

Item	Fingernail scratches it?	Penny scratches it?	Streak color	Fizz?
Eggshell or seashell				
Pencil tip				skip
Crayon			skip	skip
Bar of soap			skip	skip
Salt grains				
Outside rock				

Think like a geologist

1. Put your 6 items in order from softest to hardest.

2. Which item fizzed? Which mineral on our list is it most like?

3. Pencil 'lead' is mostly which mineral? What color was its streak?

4. Look at salt grains with a hand lens. What shape are they? Which mineral is salt?

Kitchen minerals: draw a line from the thing to its mineral

Table salt	gypsum
Pencil lead	copper
Baby powder	halite
Drywall	calcite
Chalky antacid tablets	graphite
Wires	talc

Bring to the next meeting: Bring this page to Lesson 3. Be ready to tell your group which item fizzed and why.

Lab I2: Rock Cycle Journey and Squish Lab

THIS LAB HELPS ME

- I can draw and explain the rock cycle: weathering, erosion, deposition, compaction, cementation, melting, cooling, and heat and pressure.
- I can identify the 6 metamorphic rocks (gneiss, marble, phyllite, quartzite, schist, slate), tell foliated from nonfoliated, and name each parent rock.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Rocks change from one type to another over millions of years. This is called the rock cycle. You will play it with a die and squish a model rock. These pages go behind your Rock Cycle tab.

Steps (30 minutes)

You need: 1 die, Playdough and sequins or paper bits, Colored pencils, Hand lens (optional). Set the timer for each step.

1. Me 4 min. Set a timer for 4 minutes. Read 'Rock cycle words' out loud, to yourself or to someone at home.
2. Me 5 min. Set a timer for 5 minutes. In 'Draw the rock cycle', draw boxes for magma, igneous rock, sediment, sedimentary rock, and metamorphic rock. Draw arrows between them and label each arrow with a process from the word list.
3. Me 8 min. Set a timer for 8 minutes. Start at Magma. Roll the die, find your roll in 'Dice signs', and write where you went and the process in 'My rock's journey'. Do 6 rolls. If you have a partner, start at different places and compare.
4. Me 6 min. Set a timer for 6 minutes. Squish lab: knead the sequins into the playdough and roll a ball. The flakes point every which way. Press it flat, hard, 5 times with your palm. Break it and look at the broken edge. Read 'Squish lab: foliation'.
5. Me 4 min. Set a timer for 4 minutes. Answer the 'Check yourself' questions.
6. Me 3 min. Check with the Lab I2 answer card in the Open Lab bin (or ask your Event Captain at Lesson 4). Fix mistakes in a different color. Put the playdough back in its bag.

Check your work: Use the Lab I2 answer card from the Open Lab bin, or bring the page to Lesson 4 and check it with your Event Captain. Your dice journey will be different from everyone else's; check that each process matches the sign.

Rock cycle words

Magma / lava	melted rock underground / melted rock on the surface
Cooling	magma or lava hardens into igneous rock (slow = big crystals, fast = tiny crystals or glass)
Weathering	rock breaking into small bits by water, ice, wind, or plants
Erosion	moving the bits by water, wind, or ice
Deposition	the bits settle in layers
Compaction and cementation	layers get pressed and glued into sedimentary rock
Heat and pressure	change a rock into metamorphic rock without melting it
Melting	any rock that gets hot enough becomes magma again

Dice signs

Where you are	Roll 1 or 2	Roll 3 or 4	Roll 5 or 6
Magma	cools slowly underground: go to Igneous	erupts and cools fast: go to Igneous	still hot: stay
Igneous rock	weathering and erosion: go to Sediment	heat and pressure: go to Metamorphic	melting: go to Magma
Sediment	deposition, compaction, cementation: go to Sedimentary	deposition, compaction, cementation: go to Sedimentary	still moving in a river: stay
Sedimentary rock	weathering and erosion: go to Sediment	heat and pressure: go to Metamorphic	melting: go to Magma
Metamorphic rock	weathering and erosion: go to Sediment	melting: go to Magma	more heat and pressure: stay

Draw the rock cycle

Magma, igneous, sediment, sedimentary, metamorphic. Label every arrow with a process.

My rock's journey

Roll	Where I went	Process
1		
2		
3		
4		
5		
6		

Squish lab: foliation

What happened	Pressing lines up flat flakes in layers. Heat and pressure do this to flat minerals like mica. That is foliation.
Nonfoliated	no layers; crystals fit together like sugar (marble, quartzite). Foliated rocks: slate, phyllite, schist, gneiss.

Check yourself

1. What 3 steps turn sediment into sedimentary rock?

2. Can an igneous rock become a new igneous rock? How?

3. After you pressed the playdough, how were the flakes lined up?

Bring to the next meeting: Bring this page to Lesson 4, ready to tell your rock's journey.